بسم الله الرحمن الرحيم

Ibn Sina University

Faculty of Pharmacy

First-year- Semester one

Introduction to Medical Education

Lecture 1: 15.01.2026

History of Medicine & Medical Education

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❖ Intended Learning outcomes

- **By the end of this lecture, each student will be able to:**

1. Define medicine & medical education.
2. Outline the history of medicine in different eras.
3. Outline the history of medical education in different eras.

❖ Definitions

❖ Medicine

✓The science deals with the diagnosis, treatment, and prevention of disease.

❖ Medical Education

✓The process of training individuals to become doctors or healthcare professionals.

History of medicine in different eras.

❖ Ancient Medicine, Greek Mythology – 900 BC

❖ **Aesculapius**

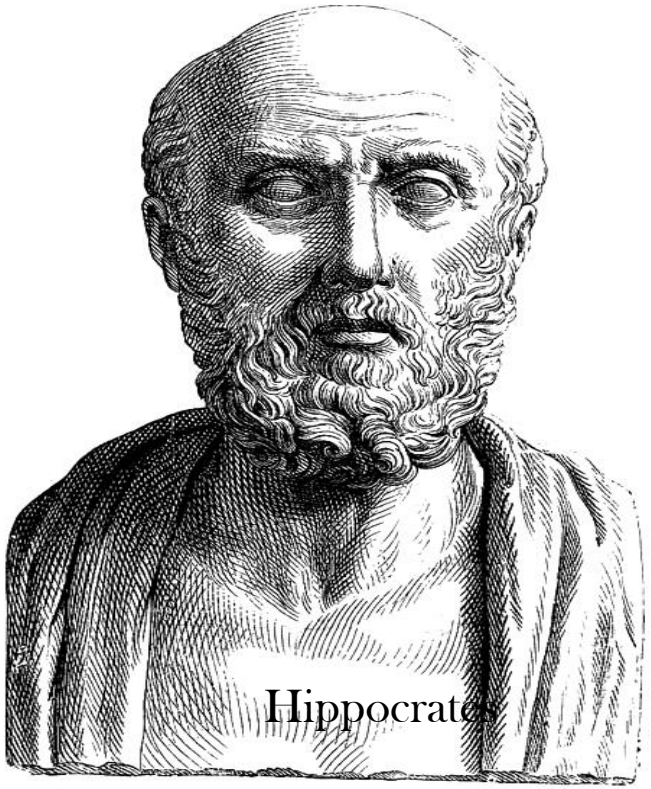
✓ Carried a rod with a single snake, which became **a medical symbol** from the fifth century BCE.

✓ **Aesculapius Snake:** its ability to shed its skin was seen as a symbol of rebirth and healing, and its poison of death.

Aesculapius

❖ Hippocrates, 500 BC (Father of Medicine)

- He collected works by many writers.
- Illness is no longer a punishment from god but **caused by natural forces**, **requiring human effort to resolve and avoid.**



❖ Hippocrates 500 BC

❖Theory of four humors (Humorism)

✓4 balanced numbers in Pythagorean philosophy: 2 pairs of opposites.

✓4 elements (earth, fire, water, air).

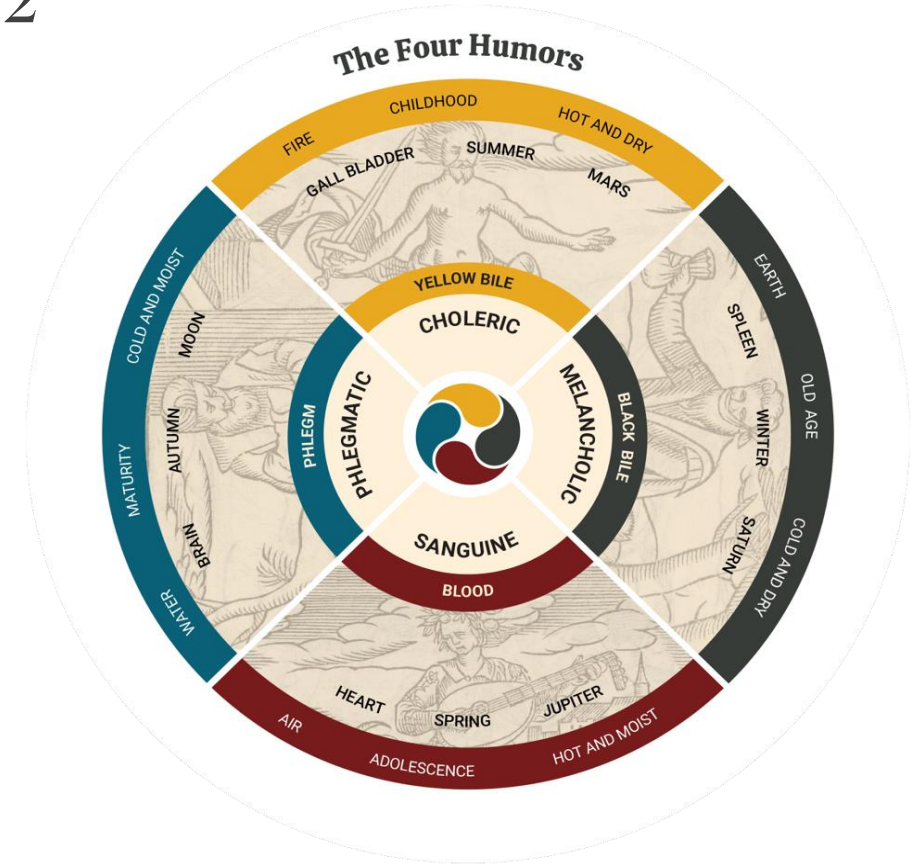
✓4 qualities (wet, dry, hot, cold).

✓4 seasons (winter, summer, autumn, spring).

✓4 body fluids (blood, phlegm, black bile, and yellow bile).

➢If 4 humors in balance = health.

➢If 4 humors are imbalanced = disease.



❖ Hippocrates 500 BC

- Holistic approach – **exercise, moderation of diet, and mental well-being.**
- Wrote on anatomy, including the **pericardium, the ventricles, and heart valves,** and the difference between veins and arteries was not discovered.
- Discussed **ethics and how a doctor should behave and present themselves.** "A physician must have **a worthy appearance; he should look healthy and be well nourished**… for most people believe that those physicians who are not tidy in their persons can not look after others well."

❖ Hippocrates 500 BC

❖ Hippocratic Oath

- "……I will soothe the pain of anyone who needs my art, and if I don't know how, I will seek the counsel of my teachers….etc.".

❖ Confidentiality

- " what I may see or hear in the course of the treatment…..which on no account must spread abroad, I will keep to myself, holding such things shameful to be spoken about "
- Considered the basis of modern medical ethics.

❖ نص قسم الاطباء:

- بسم الله الرحمن الرحيم * أقسم بالله العظيم * أن أراقب الله في مهنتي. * أن أكون على الدوام من أسباب رحمة الله، باذلاً رعايتي الطبية للقريب والبعيد، والصديق والعدو، والمحسن والمسيء والغني والفقير سواسية عندي جميع الناس من كل جنس وملة ودين وفكر. * أن أصون حياة الإنسان في كافة أطوارها، منذ أن تجمع نطفاً، باذلاً وسعي في استنقاذها من الهلاك والمرض والألم والقلق، في كل الظروف والأحوال. * أن أسعى لإتقان عملي ونماء علمي، أسخره لنفع الإنسان لا لأذاه. * أن أحفظ للمرضى كرامتهم وأستر عورتهم، وأكتم سرهم. * أن لا أشهد زوراً. * أن أعمل للارتقاء بمهنتي وتقدمها، وأن أوقر من علمني وأعلم من يصغرني، وأكون أخاً لكل طبيب يجمعني به هذا القسم. * أن يكون سلوكي مصداق إيماني، مثلاً للقيم الخالصة النقية مبرأً مما يشينني أمام الله ورسوله والمؤمنين.

❖ Aristotle 384- 322 BC, Pupil of Plato

- ✓He studied **embryology, described differences between veins and arteries, and described the aorta** (which he named) and the **course of the ureter**.
- ✓Founder of comparative anatomy.

Aristotle

❖ Galen (129-200 AD), a Greek physician

- ✓Showed that **veins were connected to the heart.**
- ✓The nerves came from the spinal cord.
- ✓Described bones and their muscle attachments.
- ✓Showed that arteries contained blood, not air.
- ✓Developed pharmaceuticals, **combining herbs and agents** into a wide variety of combinations.

❖ Islamic Medicine 700 – 1100 AD

- ✓Baghdad became a center of Islam and intellectual thought.
- ✓Greek texts **translated into Arabic.**
- ✓Developed a **pharmacy with methods of extracting and preparing medicines.**
- ✓**Reflected in Arabic are many chemical words**: alkali, alcohol, elixir, and syrup.
- ✓Understanding the **communicable nature of TB.**
- ✓Developed efficient hospitals far superior to basic prototypes made by Christians, the best known in Baghdad, Cairo, and Damascus.

❖ Ibn Sina 980–1037 (Father of modern medicine)

✓Abu Ali Al-Hussein Ibn Abdullah Ibn Sina was one of the most eminent Muslim physicians and philosophers of his day, whose influence on Islamic and European medicine persisted for centuries.

❖The Canon of Medicine (Al-Qanun fi Al-Tibb)

✓His most famous work, this comprehensive medical text, systematically organized the medical knowledge of his time. **It served as a standard reference for centuries in both the Islamic world and Europe.**

✓The Canon consists of **five books & covers a range of medical topics**, including anatomy, physiology, pathology, diagnosis, and pharmacology.

Ibn Sina

❖ Renaissance 14th century - 17th century

❖ Renaissance Surgery

- ✓ Ambroise Pare revolutionized surgery.
- ✓ For gunshot wounds, Pare tried applying clean water rather than boiling oil and found his soldiers less feverish and with less pain.
- ✓ Used **ligatures to control hemorrhage** rather than cauterizing irons.

Ambroise Pare

❖ Art and Medicine, Leonardo da Vinci 1452 - 1519

- Leonardo da Vinci made illustrations of **skeletal, muscular, venous, and nervous systems**, accurately depicted **the position of a fetus in utero**, and had unrivaled artistic ability.

Leonardo da Vinci

❖ Andreas Vesalius 1514-1564

- Discover the relationship between structures considered and structures seen as **systems rather than isolated organs.**

Andreas Vesalius

❖ Renaissance 1600s

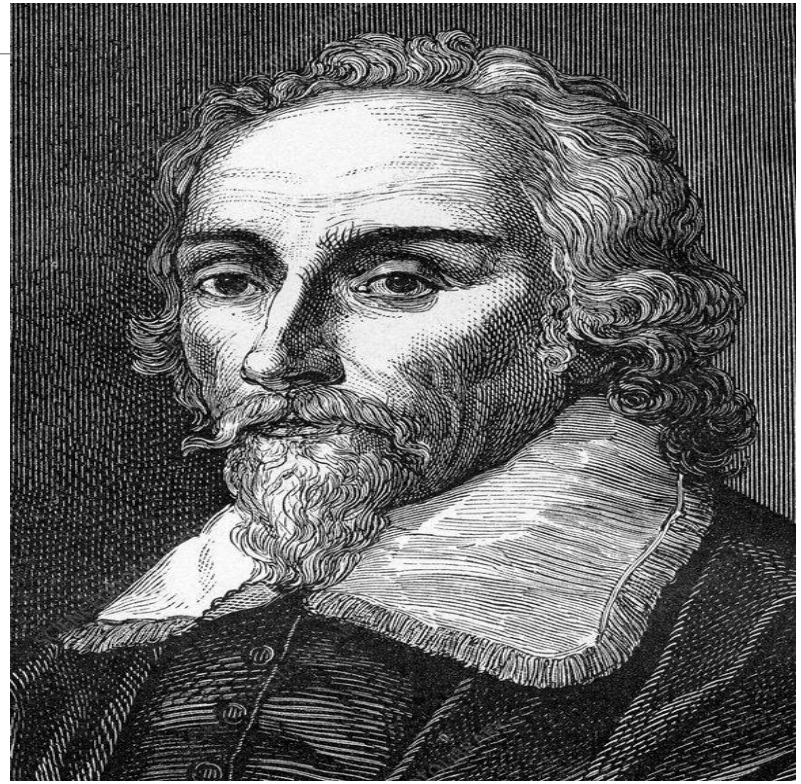
- The theory is that disease was **a distinct entity, living parasitically in the body**, as opposed to a derangement of humors normally within the body.
- Fever was not a putrefaction of humors but a **reaction to invading agents.**
- **Discovering the Microscope**..... Developed by Leeuwenhoek.

Antonie van Leeuwenhoek 1632- 1723

❖ William Harvey 1578 - 1657

- ❖ Using experiments with animals and dissection of humans, he showed that **blood flowed in one direction**, that the heart was a pump working with a relatively fixed quantity of blood, which circulated around the body, returning to the pump, to be pushed around, again and again.

- ✓ The greatest discovery of this century was **the circulation of the blood.**



William Harvey

The 18th Century - advances

- ✓William and John Hunter **developed surgery and obstetrics** and taught many distinguished physicians.
- ✓**The discovery of vaccination** by Edward Jenner, using fluid from sores on cattle, to prevent smallpox.
- ✓James Lind (1716-1764) insisted on **lemon juice to protect from scurvy**.
- ✓So the beginnings of the Industrial Revolution were in full progress.

History of medical education in different eras.

❖ History of Medical Education

- The history of medical education spans thousands of years, **evolving from apprenticeships to formalized degree programs.**

❑ **Hippocratic School (Greece, 500 BCE)**

❑ **Galen's Lectures (Rome, 100 CE):** Galen's teachings dominated Western medicine for centuries.

❑ **Middle Ages (500 - 1500 CE)**

1. Monastic Schools (Europe, 500 CE): Monasteries preserved classical medical knowledge.

2. University of Salerno (Italy, 850 CE): One of the first medical schools.

3. Scholastic Medicine (Europe, 1200 CE): Focus on theoretical medicine.

❑Renaissance and Enlightenment (1500 - 1800 CE)

1. **Anatomical Studies (1500s CE):** Andreas Vesalius's detailed anatomy texts.

2. **Royal College of Physicians (London, 1518 CE):** Established to regulate medical practice.

3. **Medical Licensing (1600s CE):** Formal licensing requirements emerged.

❖Modern Era (1800 - 2000 CE)

1. **Flexner Report (1910 CE):** Standardized medical education in the US.
2. **Medical Specialization (1900s CE):** Rise of specialized medical fields.
3. **Evidence-Based Medicine (1970s CE):** Emphasis on scientific research.
4. **Problem-Based Learning (1980s CE):** Interactive, case-based education.

❖ Contemporary Era (2000 CE - present)

1. **Competency-Based Education:** Focus on skills and outcomes.
2. **Interprofessional Education:** Collaborative training.
3. **Digital Learning:** Online resources, simulations, and virtual reality.
4. **Global Health Education:** International collaborations.
5. **Personalized Medicine:** Genomics and precision medicine.

ANY Qs

THANKS

HAVE A NICE DAY....

Next:... Value of studying medicine and its importance & Responsibilities of medical practitioners.

SUCCESS IS NO ACCIDENT.
It is hard work, perseverance,
learning, studying, sacrifice
and most of all, **love** of what you are doing.
- Pele